Scored Consistency Distillation of SD3.5-Medium: Selecting Teacher Trajectories with a Reference Photograph

Technical report

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Abstract

We distil Stable Diffusion 3.5 Medium into a 4-step student with consistency distillation. For every training caption the frozen teacher generates four candidate trajectories from four seeds. A scorer picks one of the four, and the student is trained on that trajectory only. The baseline picks at random. The main scorer is the cosine similarity between DINOv2 mean-pooled patch features of the candidate image and of the real photograph that the caption describes. It uses no text model. On 3,000 COCO captions, three seeds, and 6,000 updates, this selection improves T2I-CompBench by +0.014 and GenEval2 by +2.2 points over random selection, with 95% bootstrap intervals that exclude zero, and does not cost fidelity. At 118k captions and 57k updates, three training seeds, with weight averaging of the last five checkpoints, the gain is +0.0175 CompBench ($p = 10^{-14}$) and +1.3 GenEval2; it is +0.013 under the official ten-images-per-prompt CompBench protocol ($p = 10^{-41}$), and the scored student has better CMMD, precision and recall on every seed. The gain is unchanged by the optimizer batch size, does not grow with eight candidates instead of four, and vanishes with a stronger teacher. Argmax is the right selection rule: exact Boltzmann weighting of all four candidates ties it at 2.3 times the compute, and drawing one candidate from the same weights loses most of the gain through target churn. The same pipeline with VQAScore as the scorer gives similar gains but shares a model family with the evaluators. This document gives the full recipe: data, model, scorers, loss, hyperparameters, evaluation protocol, results, ablations and qualitative examples.

Contents

1 Summary

Method in one paragraph. The teacher is SD3.5-Medium with classifier-free guidance $w = 7$ and 8 Euler steps. For each caption we roll out $N = 4$ candidates from four fixed seeds. A scorer ranks the four decoded images and the best one is stored in a manifest. Training re-rolls only the chosen trajectory and regresses the student's clean-image estimate at a noisier state onto the teacher's clean-image estimate at a less noisy state. The student is sampled with 4 steps and no guidance. The only thing that differs between arms is which one of the four trajectories the student sees.

Main numbers. Table 1 gives the headline results. All deltas are paired against the random-selection student on identical prompts.

Table 1: Headline results. Alignment deltas are against the random-selection student. The 3k rows use three training seeds; intervals are 95% hierarchical bootstrap over seeds and prompts. The 118k rows use three training seeds and weight-averaged checkpoints, with per-prompt paired tests pooled over seeds; the VQAScore arm at 118k is one seed. CMMD is the seed mean.

Setting	Scorer	CompBench Δ	GenEval2 Δ (×100)	CMMD ↓
3k captions, 6k updates	random (baseline)	0.4584	20.73	0.963
3k captions, 6k updates	DINOv2 patches	+0.0140 [+0.0051, +0.0237]	+2.17 [+0.24, +4.08]	0.893
3k captions, 6k updates	VQAScore	+0.0233 [+0.0097, +0.0364]	+2.34 [+0.55, +4.05]	0.883
118k captions, 57k updates, 3 seeds	random (baseline)	0.4668	20.29	0.84
118k captions, 57k updates, 3 seeds	DINOv2 patches	+0.0175 ($p = 10^{-14}$)	+1.29 ($p = 0.02$)	0.78
118k, official 10-image CompBench, 3 seeds	DINOv2 patches	+0.0131 ($p = 2 \times 10^{-41}$)	–	–
118k captions, 57k updates, 1 seed	VQAScore	+0.0185 ($p = 10^{-7}$)	+3.05 ($p = 0.001$)	0.79
Teacher, 28 steps, $w = 7$	–	0.5053	17.05	0.64

2 Base model

2.1 SD3.5-Medium as a rectified flow

SD3.5-Medium is a 2.5B-parameter Multimodal Diffusion Transformer (MMDiT). We use it unchanged. It generates in the latent space of its VAE. At 512 × 512 pixels a latent is $z \in \mathbb{R}^{16\times 64\times 64}$, so $D = 65{,}536$.

Rectified flow defines a straight path between a clean latent x_0 and Gaussian noise $\varepsilon \sim \mathcal{N}(0, I)$:

$$z_\sigma = (1 - \sigma)\, x_0 + \sigma\, \varepsilon, \qquad \sigma \in [0, 1]. \tag{1}$$

The network predicts the velocity $v = \varepsilon - x_0$. Sampling is Euler integration down a σ grid:

$$z_{k+1} = z_k + (\sigma_{k+1} - \sigma_k)\, v_\theta(z_k, \sigma_k, c). \tag{2}$$

The network is conditioned on the timestep $t = 1000\,\sigma$. The grid is built by the scheduler with a shift $\lambda = 3$: $\sigma \mapsto \lambda\sigma/(1 + (\lambda - 1)\sigma)$. The grids used in this work are given in Table 2.

Table 2: σ grids. The 4-step grid is not a subset of the 8-step grid; only the endpoints coincide. It is a subset of the 28-step grid (every ninth state).

Grid	σ values
8 steps (teacher, training)	1, 0.948, 0.883, 0.801, 0.694, 0.548, 0.338, 0.009, 0
4 steps (student, inference)	1, 0.858, 0.602, 0.009, 0
28 steps (teacher reference)	1 down to 0, same shift, 28 intervals

2.2 Conditioning

The caption enters through three frozen text encoders: CLIP-L and CLIP-G as token sequences and a pooled 2048-d vector, and T5-XXL as a token sequence. The teacher uses classifier-free guidance:

$$v^{\mathrm{cfg}} = v_\theta(z, \sigma, \varnothing) + w\big(v_\theta(z, \sigma, c) - v_\theta(z, \sigma, \varnothing)\big), \qquad w = 7. \tag{3}$$

The VAE is frozen and is used only to decode candidates for scoring and to decode images for evaluation.

3 Data

3.1 Training captions and reference photographs

Training captions come from COCO `train2017`. Each caption is paired with the photograph it was written for. A caption string that COCO attaches to more than one photograph is kept only for its first photograph, so every caption maps to exactly one reference. Two pools are used:

- **3k pool**: 3,000 captions drawn at random with a fixed seed. Used for every three-seed experiment and every ablation.
- **118k pool**: 113,948 captions (named after its initial count of 118,287 before deduplication and reference checks). Used for the single-seed scale experiment.

The reference photograph is only used by the scorer. It never enters training.

3.2 Evaluation pools

All evaluation uses one sealed pool built once with a fixed seed (Table 3). Evaluation references and fidelity captions come from COCO `val2017`, so no training caption or photograph appears in any evaluation set.

Table 3: Evaluation pools. Every arm and every seed is evaluated on exactly these prompts.

Pool	Prompts	Source
T2I-CompBench	2,398	300 per category (298 for 2D-spatial), 8 categories, official prompt files
GenEval2	800	official GenEval2 prompts, 3 to 10 atoms each, 100 per atom count
COCO VQAScore (in-domain)	5,000	`val2017` captions, one per image
Fidelity	5,000	`val2017` captions; reals are the 5,000 matching photographs

4 Candidate pool

For each caption we draw $N = 4$ candidates. Candidate j of caption `idx` uses the seed $1000 \cdot \texttt{idx} + j$, so the whole pool is reproducible from integers. Each candidate is an 8-step Euler rollout of the teacher at $w = 7$ and 512×512:

$$\varepsilon^{(j)} \sim \mathcal{N}(0, I) \text{ seeded by } 1000 \cdot \texttt{idx} + j, \qquad \hat{x}_0^{(j)} = \mathrm{Euler}_{K=8}\big(\varepsilon^{(j)};\, v^{\mathrm{cfg}}, c\big). \tag{4}$$

Each candidate is decoded to an image $I_j = \mathrm{Dec}(\hat{x}_0^{(j)})$ for scoring. The four candidates differ only by their noise seed. The training rollout in Section 6 uses the same sampler and the same seed, so the cached candidate and the training trajectory are the same object.

Nothing is stored except the seed, the scores and the chosen index. Trajectories are re-rolled from the seed during training.

5 Scorers

A scorer assigns a number S_j to each candidate. Selection is hard:

$$j^\star = \arg\max_{j \in \{1,\dots,4\}} S_j. \tag{5}$$

The index $j^\star$ is computed once, offline, and stored in a manifest keyed by caption. No gradient ever flows through a scorer. Section 10 compares this rule with Boltzmann-weighted and sampled selection; argmax is the best of them. The random baseline replaces $j^\star$ by a uniform draw that is also stored in the manifest and is therefore identical across training seeds.

5.1 DINOv2 mean-pooled patches (main scorer)

We use DINOv2-B (`facebook/dinov2-base`, $d = 768$) with its standard preprocessing: resize the short side to 256, centre-crop to 224×224, ImageNet normalisation. The model returns one CLS token and $P = 256$ patch tokens $h_1, \dots, h_P$ (a 16×16 grid, patch size 14). We drop the CLS token, average the patch tokens and normalise:

$$u^{\text{pat}}(I) = \frac{\frac{1}{P}\sum_{p=1}^{P} h_p}{\left\|\frac{1}{P}\sum_{p=1}^{P} h_p\right\|}. \tag{6}$$

The score is the cosine similarity to the reference photograph x^{ref}:

$$S_j = \left\langle u^{\text{pat}}(I_j),\, u^{\text{pat}}(x^{\text{ref}})\right\rangle. \tag{7}$$

Candidates are decoded at 512×512. The reference photograph is first resized (squashed, aspect ratio not preserved) to 512×512 with bicubic interpolation, so both enter the DINO processor as squares and the centre crop discards nothing. There is no text model in this scorer. The caption enters only through its photograph.

5.2 Other scorers

- **DINOv2 CLS.** Same model and preprocessing, but the normalised CLS token $h_0/\|h_0\|$ instead of the patch mean.
- **CLIP-I.** Cosine between CLIP ViT-L/14 image embeddings of the candidate and of the reference photograph.
- **VQAScore.** The probability that CLIP-FlanT5-XXL (11B) answers "yes" to "Does this figure show *caption*?". It reads the caption and needs no photograph. It shares a model family with the BLIP-VQA part of CompBench and the VLM judge of GenEval2, so its gains are partly circular. It is the reference scorer, not the proposed one.
- **ImageReward.** A BLIP-based learned human-preference model that reads the caption.
- **PickScore.** A CLIP-H model fine-tuned on human preferences that reads the caption.

5.3 How well each scorer selects

Table 4 measures selection quality offline on the 3k pool. The ground truth is VQAScore. The random pick scores 0.8429 on average and the best-of-four oracle 0.9334, so the headroom is 0.0905. The table gives the fraction of that headroom each scorer recovers and how often its pick is the oracle pick (chance is 25%).

The patch scorer and the CLS scorer agree on only 54.5% of picks, so they are different selectors. ImageReward and PickScore agree with each other on 50.0% of picks and with VQAScore on 42.7% and 41.5%.

Table 4: Offline selection quality on the 3k pool, 12,000 candidates. Headroom is measured against the VQAScore best-of-four oracle, so the caption-reading scorers have an advantage here that is partly shared with the ground truth.

Scorer	Needs	VQAScore headroom recovered	Top-1 agrees with oracle
VQAScore (oracle)	caption, 11B VLM	100%	1.000
ImageReward	caption, BLIP	64.5%	0.427
PickScore	caption, CLIP-H	55.6%	0.415
DINOv2 patches	reference photo	43.6%	0.356
CLIP-I	reference photo	41.4%	0.328
DINOv2 CLS	reference photo	39.9%	0.324

6 Distillation

6.1 Teacher and student

The teacher is the pretrained SD3.5-Medium transformer with frozen weights θ_T, kept in bf16 with no gradient, always evaluated with guidance $w = 7$. It is never updated. There is no exponential moving average, no momentum target and no stop-gradient copy of the student. The student v_θ is a full copy of the same transformer, initialised from θ_T, and is the only network that trains. Every target is a deterministic function of the frozen teacher's own trajectory, so it does not depend on θ and is identical across epochs.

6.2 Trajectory and targets

For each training caption we roll out only the selected candidate with the teacher, $w = 7$, $K = 8$ steps, giving $z_0, z_1, \ldots, z_8$. The supervised states are the scoring window $[0.4, 0.9]$ of the grid, $\mathcal{K} = \{3, 4, 5, 6, 7\}$, so $K_w = 5$ states per update.

For each $k \in \mathcal{K}$ the teacher's realised chord across its own step is

$$\bar{v}_k = \frac{z_{k+1} - z_k}{\sigma_{k+1} - \sigma_k}, \tag{8}$$

and the clean-latent estimate implied by that chord is

$$\tilde{x}_0^{(k)} = z_k - \sigma_k \, \bar{v}_k. \tag{9}$$

This is exact algebra for a straight path: substituting $z_k = (1 - \sigma_k)x_0 + \sigma_k\varepsilon$ and $\bar{v} = \varepsilon - x_0$ recovers x_0.

6.3 Student input and loss

The student is placed at a noisier state. For each k we draw $\Delta_k \sim \mathcal{U}\{1, 2, 3\}$, independently per state and afresh on every visit, and set $s = k - \Delta_k$. The five inputs $z_{k-\Delta_k}$ are batched into one forward pass. The student uses one conditional pass, no guidance, and forms its own clean-latent estimate:

$$\hat{x}_0^{(s)} = z_s - \sigma_s \, v_\theta(z_s, \sigma_s, c). \tag{10}$$

The loss is a pseudo-Huber distance in x_0 space, averaged over the window:

$$\mathcal{L}(\theta) = \frac{1}{K_w} \sum_{k \in \mathcal{K}} \left[\sqrt{\left\| \hat{x}_0^{(k-\Delta_k)} - \mathrm{sg}\left[\tilde{x}_0^{(k)}\right] \right\|_2^2 + c^2} - c \right], \qquad c = 0.00054\sqrt{D} \approx 0.138. \tag{11}$$

The norm is summed over all D latent dimensions, so this is one distance per sample. sg[·] is stop-gradient. The student stands at a noisier point and must name the clean image that the teacher only reaches from a less noisy point. Composing these skips is what turns 8 teacher steps into 4 student steps.

Because the student trains with a single conditional pass against $w = 7$ teacher trajectories, guidance is absorbed into its conditional prediction. The student must be sampled at $w = 1$. Sampling it at $w = 7$ applies guidance twice and roughly halves its scores.

6.4 Training details

Table 5 lists both training configurations. Each optimiser step processes one caption per GPU: one teacher rollout of the selected candidate (16 teacher forward passes, 8 steps with a conditional and an unconditional branch) and the five batched student passes. The training seed controls only the Δ_k draws and the caption order. The selection index is fixed in the manifest.

Table 5: Training configurations. Both use the same frozen trainer code.

	3k configuration	118k configuration
Captions	3,000	113,948
GPUs (H200)	1	4, DDP, one caption per GPU per step
Updates	6,000 (2 passes)	56,974 (2 passes)
Seeds	3	3 (VQAScore arm: 1)
Optimiser	AdamW, $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$	same
Weight decay	0	0
Learning rate	10^{-5}	2×10^{-5}
Warm-up	300 steps, linear, then constant	1,000 steps, linear, then constant
Gradient clip	global norm 1.0	global norm 1.0
Precision	fp32 master weights, bf16 autocast	same
Gradient checkpointing	on	on
Teacher, text encoders, VAE	frozen, bf16	same
Captions per update	1	4 (one per GPU; 16 with accumulation, Section 9.5)
Reported weights	final step	uniform average of steps 40k, 45k, 50k, 55k, final
Wall-clock per run	1.6 h	12.5 h
Peak GPU memory	–	66 GB per GPU

The raw gradient norm of this loss is of order 10^2 throughout training, so the clip is active on every step. Every update is therefore a normalised-gradient step of size η. All arms share this, so it does not affect comparisons, but the learning rate and the clip threshold are not separately tunable.

6.5 Weight averaging at 118k

The 118k runs save a checkpoint every 5,000 steps. Consecutive checkpoints of one run differ by up to 0.035 CompBench (Section 9). The reported 118k student is the uniform average of the last five checkpoints (steps 40k, 45k, 50k, 55k and 56,974). Averaging is applied identically to every arm.

6.6 Sampling

The student is sampled with 4 Euler steps on the 4-step grid of Table 2, $w = 1$, at 512×512. Every number in this document for a distilled student comes from this setting.

7 Evaluation

7.1 Alignment

- **T2I-CompBench.** 2,398 prompts, official scorers: BLIP-VQA for color, shape and texture; UniDet for 2D-spatial, 3D-spatial and numeracy; CLIPScore for non-spatial; the

3-in-1 score for complex. The overall score is the unweighted mean of the eight category means. One image per prompt, seed equal to the prompt index, so every arm sees identical noise.

- **GenEval2.** 800 prompts. Each prompt is decomposed into 3 to 10 atoms tagged object, count, attribute, position or verb. Qwen3-VL scores each atom with a probability (Soft-TIFA). The prompt score is the geometric mean of its atom scores, and the official number is the mean prompt score $\times 100$.
- **COCO VQAScore.** 5,000 held-out `val2017` captions scored by VQAScore. This is in-domain for the training captions. For the VQAScore-selected arm this metric is circular.

7.2 Fidelity

5,000 images from `val2017` captions against the 5,000 matching real photographs, centre-cropped to squares. FID uses clean-fid. CMMD uses the reference implementation: CLIP ViT-L/14 at 336 px, centre crop then bicubic resize, unit-normalised embeddings, RBF kernel with $\sigma = 10$, biased estimator, $\times 1000$. Precision and recall use k-NN manifolds. CMMD intervals are 1,000-sample bootstraps.

7.3 Statistics

Deltas are paired on identical prompts. For the three-seed experiments on the 3k pool, the interval is a hierarchical bootstrap that resamples seeds and prompts jointly (4,000 resamples). The 118k and selection-rule contrasts report per-prompt paired t-tests (Wilcoxon tests agree throughout), pooled over seeds where there are several, with the per-seed deltas alongside.

7.4 Two properties of the metrics that matter for reading the tables

GenEval2 rewards partial credit. On the geometric-mean score the 28-step teacher (17.05) scores below every 4-step student (20.7 to 23.7). On strict accuracy, the share of prompts with every atom above 0.5, the teacher leads (10.2% against 6.7 to 8.2%), and it leads on the arithmetic atom mean, on CompBench and on COCO VQAScore. The reason is that the judge gives the teacher's wrong atoms probabilities near zero, while the students' wrong atoms, mostly attribute and position atoms, get mid-range probabilities. One near-zero atom sends a geometric mean to zero, so 61% of the teacher's prompts collapse against 35 to 39% for the students, although both fail about 2.3 atoms per prompt. GenEval2 scores should therefore be read as relative numbers between students, not as evidence that a student beats the teacher.

CompBench 2D-spatial has dead prompts. The official 2D-spatial evaluator matches prompt nouns to detector labels by exact string equality. 31.8% of the official spatial prompts, and 98 of our 298, contain a noun that no label matches (for example *cow*, *sofa*, *wallet*), and they score zero for every possible image. This scales the spatial column by a constant and does not reorder models (rank correlation 0.91 between official and repaired scoring, no significant reversal). Spatial deltas are therefore understated by about $1.5\times$ relative to other categories.

8 Results on the 3k pool

8.1 Alignment, all scorers

Table 6 is the main result. Every arm is three seeds on the same 3k pool, the same candidate cache and the same configuration.

Three points follow from the table.

Table 6: 3k pool, 6,000 updates, three seeds. Absolute scores are seed means. Deltas are against the random-selection student with 95% hierarchical bootstrap intervals. Bold intervals exclude zero.

Scorer	Needs	CompBench	CompBench Δ	GenEval2	GenEval2 Δ (×100)	COCO VQAScore
random (baseline)	–	0.4584	–	20.73	–	0.8733
DINOv2 patches	photo	0.4725	**+0.0140 [+0.0051, +0.0237]**	22.89	**+2.17 [+0.24, +4.08]**	0.8850
DINOv2 CLS	photo	0.4620	+0.0036 [−0.0062, +0.0142]	21.73	+1.00 [−0.80, +2.86]	0.8799
CLIP-I	photo	0.4595	+0.0010 [−0.0112, +0.0130]	21.23	+0.51 [−1.44, +2.49]	0.8748
VQAScore	caption	0.4817	**+0.0233 [+0.0097, +0.0364]**	23.06	**+2.34 [+0.55, +4.05]**	0.8873
ImageReward	caption	0.4806	**+0.0221 [+0.0127, +0.0322]**	22.17	+1.45 [−0.43, +3.37]	0.8854
PickScore	caption	0.4713	**+0.0128 [+0.0012, +0.0245]**	22.52	+1.80 [−0.05, +3.72]	0.8850
Teacher, 8 steps, $w = 7$ (source)		0.4544		17.58		–
Teacher, 28 steps, $w = 7$		0.5053		17.05		0.9102
Base model, 4 steps, $w = 7$		0.2557		7.32		0.4762

- The DINOv2 patch scorer, which reads no text, recovers 60% of VQAScore's CompBench gain and 93% of its GenEval2 gain.
- Mean-pooled patches beat the CLS token. Head-to-head on identical prompts the patch arm is ahead of the CLS arm by +0.0104 CompBench ($p = 10^{-5}$), +0.0108 on UniDet, +0.0154 on BLIP-VQA and +1.17 GenEval2, positive on all three seeds for every endpoint. CLIP-I, the other reference-photo scorer, is a null.
- Every 4-step student beats the 8-step teacher it was distilled from on CompBench. The 28-step teacher remains clearly better on CompBench and COCO VQAScore. Its lower GenEval2 number is the metric property of Section 7.4.

8.2 Fidelity

Table 7 shows that scored selection does not cost fidelity. Every scored arm has lower CMMD than the random baseline, and precision and recall are equal or higher. FID differences between students are within the split-half spread of about 0.5 and should not be read.

Table 7: Fidelity on 5,000 COCO `val2017` captions, 4 steps, $w = 1$, seed means over three runs. CMMD bootstrap intervals are about ±0.01.

Arm	FID ↓	CMMD ↓	Precision ↑	Recall ↑
random (baseline)	32.64	0.963	0.440	0.041
DINOv2 patches	30.72	0.893	0.465	0.052
DINOv2 CLS	30.98	0.913	0.467	0.046
CLIP-I	32.45	0.870	0.484	0.049
VQAScore	30.97	0.883	0.461	0.052
ImageReward	30.71	0.867	0.470	0.056
PickScore	31.02	0.907	0.444	0.049
Teacher, 28 steps, $w = 7$	29.21	0.640	0.549	0.156
Base model, 4 steps, $w = 7$	94.48	2.150	0.178	0.006

8.3 Absolute scores per category

Tables 8 to 10 give per-category absolute scores for the teacher, the base model and the four main arms. The largest per-skill gain of the DINOv2 patch scorer on GenEval2 is on *position* (51.4 against 45.9), consistent with mean-pooled patches carrying layout, and its advantage grows with prompt complexity.

Table 8: GenEval2 (Soft-TIFA, Qwen3-VL judge), $\times 100$, on the sealed 800-prompt pool. *Overall* is the official score, the mean over prompts of the geometric mean of a prompt's atom scores ($\pm$ standard deviation over training seeds). Skill columns are the mean score of the atoms tagged with that skill. *Atom mean* is the arithmetic mean over all atoms. *Collapsed* is the share of prompts whose geometric mean falls below 0.05. Teacher and base rows are single deterministic runs. Bold marks the best 4-step distilled arm in each column.

Model	Steps / CFG	Seeds	Overall	Object	Count	Attribute	Position	Verb	Atom mean	Collapsed (%)
Teacher, 28 steps, CFG 7	28 / 7	1	17.05	87.6	44.7	66.1	45.7	17.4	64.7	60.6
Teacher, 8 steps, CFG 7 (source)	8 / 7	1	17.58	83.7	40.6	67.5	42.5	18.3	62.1	54.8
Base, 4 steps, CFG 7	4 / 7	1	7.32	43.2	20.8	49.9	26.9	4.4	36.2	79.0
Naive CD (random pick)	4 / 1	3	20.73 ± 0.94	80.5	34.8	69.5	45.9	21.4	59.8	38.5
Scored CD, DINO CLS	4 / 1	3	21.73 ± 0.76	82.7	34.4	70.1	48.5	24.2	60.9	37.0
Scored CD, DINO patches	4 / 1	3	22.89 ± 1.01	82.8	34.6	70.5	**51.4**	26.0	61.4	35.4
Scored CD, VQAScore	4 / 1	3	$\mathbf{23.06 \pm 0.15}$	**84.3**	**35.6**	**70.7**	47.8	**26.6**	**62.1**	**34.8**

Table 9: GenEval2 by prompt complexity: mean prompt score ($\times 100$) in each atom-count bucket, 100 prompts per bucket, averaged over training seeds where applicable. Bold marks the best 4-step distilled arm in each column.

Model	3	4	5	6	7	8	9	10
Teacher, 28 steps, CFG 7	36.3	32.5	22.5	19.9	12.8	4.7	5.5	2.3
Teacher, 8 steps, CFG 7 (source)	34.2	35.6	23.2	18.1	11.2	9.0	4.9	4.2
Base, 4 steps, CFG 7	21.0	8.3	8.7	6.9	3.9	4.7	3.0	2.1
Naive CD (random pick)	38.9	34.2	22.8	20.9	15.9	13.6	11.7	7.8
Scored CD, DINO CLS	43.2	34.2	24.4	22.2	15.5	13.7	13.0	7.5
Scored CD, DINO patches	41.6	**35.6**	**25.4**	**24.7**	**17.7**	**15.3**	**14.3**	8.6
Scored CD, VQAScore	**48.4**	33.4	24.6	24.1	17.3	14.0	13.2	**9.4**

9 Results at 118k captions

9.1 Weight-averaged students, three seeds

Table 11 gives the 118k result. Every arm is three training seeds on the same cache and the same configuration, and the reported weights are the uniform average of the last five checkpoints of each run. This is the largest and best-performing setting in this work.

The per-seed DINOv2 patch gains are +0.022, +0.011 and +0.019 CompBench (standard deviation 0.006, so a single-seed contrast on this branch resolves about 0.01). The gain is present in both evaluator families of CompBench: UniDet +0.026 ($p = 5 \times 10^{-9}$) and BLIP-VQA +0.017 ($p = 10^{-5}$); the CLIPScore category is a null (+0.0002). The averaged patch student is 0.021 CompBench below the 28-step teacher and 0.030 above its 8-step source.

9.2 Official protocol: ten images per prompt

T2I-CompBench's official protocol scores ten images per prompt. Table 12 repeats the evaluation of the same seven averaged models under it (images seeded by prompt and sample index, identical across arms). The gain holds on every seed with the same category profile (Table 13, Figure 1). Its size is about three quarters of the one-image number, because ten images per prompt is a less noisy estimate for both arms, and its statistical strength is far higher.

9.3 Fidelity, three seeds

Table 14 gives fidelity per seed. The 0.05 to 0.06 CMMD advantage of the scored student holds on every seed, far outside the ± 0.01 bootstrap interval, and so do precision and recall. FID is

Table 10: T2I-CompBench on the sealed 2,398-prompt pool, official scorers, one image per prompt. *Mean* is the unweighted mean of the eight categories (± standard deviation over training seeds); category columns are averaged over seeds. Bold marks the best 4-step distilled arm in each column.

Model	Mean	Color	Shape	Texture	Spatial	3D-spatial	Numeracy	Non-spatial	Complex
Teacher, 28 steps, CFG 7	0.5053	0.7977	0.5732	0.7427	0.2716	0.3662	0.5915	0.3154	0.3837
Teacher, 8 steps, CFG 7 (source)	0.4544	0.7142	0.4988	0.6750	0.2425	0.3276	0.5518	0.3093	0.3163
Base, 4 steps, CFG 7	0.2557	0.4168	0.2921	0.4053	0.0582	0.1406	0.2574	0.2707	0.2048
Naive CD (random pick)	0.4584 ± 0.0075	0.7840	0.5435	0.7081	0.1783	0.2806	0.5015	0.3080	0.3634
Scored CD, DINO CLS	0.4620 ± 0.0046	0.7904	0.5388	0.6983	0.1878	0.2830	0.5205	0.3088	0.3685
Scored CD, DINO patches	0.4725 ± 0.0014	0.8041	0.5525	**0.7172**	0.2061	0.2839	0.5339	0.3093	0.3727
Scored CD, VQAScore	**0.4817 ± 0.0106**	**0.8120**	**0.5651**	0.7157	**0.2332**	**0.3063**	**0.5366**	**0.3096**	**0.3754**

Table 11: 118k captions, 56,974 updates on 4 GPUs, three training seeds, uniform average of the last five checkpoints. Per-seed scores and seed means; deltas are per-prompt paired tests against the random-selection student, pooled over seeds. The VQAScore arm has one seed and is paired against seed 0 of the baseline.

Scorer	CompBench (s0 / s1 / s2)	mean	Δ (p)	GenEval2 (s0 / s1 / s2)	mean	Δ (p)
random (baseline)	0.4642 / 0.4709 / 0.4653	0.4668	–	20.61 / 20.52 / 19.74	20.29	–
DINOv2 patches	0.4865 / 0.4821 / 0.4843	**0.4843**	+0.0175 (10^{-14})	22.05 / 21.22 / 21.48	21.58	+1.29 (0.02)
VQAScore (seed 0)	0.4827	0.4827	+0.0185 (10^{-7})	23.66	**23.66**	+3.05 (0.001)

within its split-half spread.

9.4 Why averaging is needed

Table 15 and Figure 2 evaluate every saved checkpoint of the three seed-0 runs. Consecutive checkpoints of one run differ by up to 0.035 CompBench, which is larger than the effect. The cause is the fixed-size normalised step (Section 6) with no averaging: the weights move on a plateau and each checkpoint samples a different point of it. Paired across the eight checkpoints, the patch arm is ahead of random on CompBench by +0.014 on average (7 of 8 checkpoints, $p = 0.02$) and the VQAScore arm by +0.010 (6 of 8, $p = 0.05$). Averaging the last five checkpoints raised every arm and moved the scored arms most. Over the three seeds, the raw final checkpoints give a patch-minus-random gap of only +0.007 (+0.001, +0.003, +0.017; $p = 0.003$) against +0.0175 for the averaged weights, so averaging is not optional at this scale, and it is applied identically to every arm.

The averaging also improved fidelity for every arm at seed 0: CMMD went from 0.87 to 0.84 for random, 0.86 to 0.78 for the patch arm and 0.87 to 0.79 for the VQAScore arm.

9.5 Optimizer batch size

Every update above processes four captions, one per GPU. To test whether the oscillation or the gap depends on that, both arms were retrained at seed 0 with gradient accumulation over four captions per GPU: 16 captions per update, 14,244 updates for the same two passes over the data, checkpoints every 1,250 updates so that the last-five average spans the same data window. Nothing else changed. Table 16 gives the result. The selection gap is unchanged (+0.0221 against +0.0223 for this seed at batch 4). Batch 16 lifts both arms by the same +0.005, which is not significant at one seed. It removes the oscillation of the random arm (raw final and average within 0.001) but not of the scored arm (0.012 apart), so averaging remains necessary, and batch 4 with averaging remains the reported configuration. This row is a robustness check.

Table 12: Official ten-images-per-prompt T2I-CompBench on the 118k weight-averaged models, per training seed. Paired deltas against the random-selection student.

Scorer	seed 0	seed 1	seed 2	mean, Δ (p)
random (baseline)	0.4704	0.4723	0.4729	0.4719
DINOv2 patches	0.4878	0.4838	0.4832	0.4850, +0.0131 (2×10^{-41})
VQAScore	0.4835	–	–	+0.0131 (3×10^{-18})

Table 13: CompBench per category at 118k, three-seed means of the weight-averaged students, under the one-image and the official ten-image protocol.

	one image per prompt			ten images per prompt (official)		
Category	random	DINOv2 patches	Δ	random	DINOv2 patches	Δ
color	0.7755	0.7978	+0.0223	0.7747	0.7950	+0.0203
shape	0.5338	0.5455	+0.0118	0.5367	0.5575	+0.0207
texture	0.7051	0.7233	+0.0182	0.7166	0.7260	+0.0094
2D-spatial	0.1971	0.2279	+0.0308	0.2112	0.2284	+0.0172
3D-spatial	0.2999	0.3224	+0.0224	0.3116	0.3252	+0.0136
numeracy	0.5483	0.5726	+0.0242	0.5498	0.5619	+0.0122
non-spatial	0.3141	0.3142	+0.0002	0.3137	0.3139	+0.0002
complex	0.3606	0.3707	+0.0101	0.3607	0.3717	+0.0111
mean	0.4668	0.4843	+0.0175	0.4719	0.4850	+0.0131

9.6 Number of candidates

The cache was extended from four to eight candidates per caption with the same seed convention (candidate j from seed $1000 \cdot \texttt{idx} + j$, $j = 4, \ldots, 7$), keeping the random arm's stored draw, so the $N = 4$ arms remain the exact control. Offline, best-of-eight raises the VQAScore oracle on this pool from 0.9335 to 0.9487 (headroom over the random pick +0.094 → +0.109, 16% more), and the DINOv2-selected candidate from 0.8796 to 0.8862, the same 43% of the respective headroom. Trained with argmax over eight candidates, three seeds, the averaged students score 0.4862 / 0.4859 / 0.4838 CompBench (mean 0.4853) and 21.68 / 21.44 / 21.50 GenEval2 (21.54): +0.0010 ($p = 0.64$) and −0.04 against the $N = 4$ arm, a null. Both arms are about +0.018 over random. Four candidates saturate this scorer on this teacher; the extra offline headroom of eight does not convert into a trained gain.

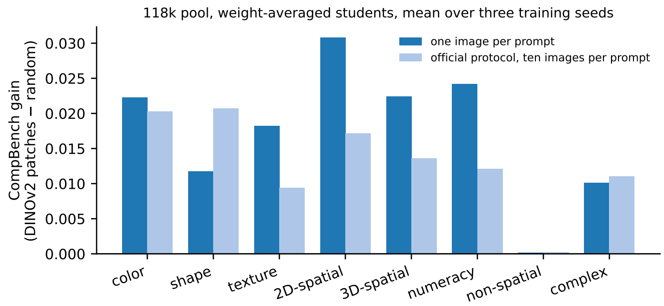

Figure 1: Per-category CompBench gain of DINOv2 patch selection over random selection at 118k (three-seed means of the averaged students) under both protocols. The gain is spread over the attribute-binding, spatial and counting categories and is zero on non-spatial, whose CLIPScore evaluator does not separate any two students in this work.

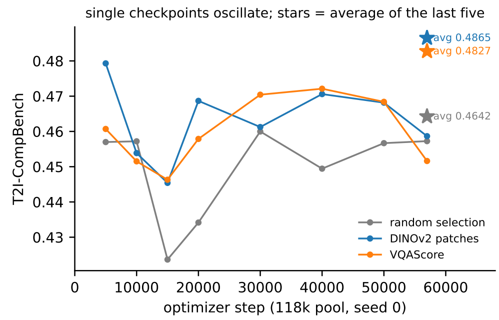

Figure 2: CompBench of every saved checkpoint of the seed-0 118k runs; the star marks the weight average of the last five checkpoints of each arm (Section 9.4).

Table 14: Fidelity at 118k, weight-averaged students, 5,000 COCO `val2017` captions, 4 steps, $w = 1$, per training seed (s0 / s1 / s2).

Arm	FID ↓	CMMD ↓	Precision ↑	Recall ↑
random (baseline)	31.08 / 30.97 / 31.44	0.84 / 0.84 / 0.83	0.485 / 0.489 / 0.503	0.059 / 0.066 / 0.062
DINOv2 patches	30.98 / 30.83 / 31.47	**0.78 / 0.79 / 0.78**	**0.522 / 0.511 / 0.522**	**0.071 / 0.068 / 0.063**
VQAScore (seed 0)	30.45	0.79	0.510	0.061
Teacher, 28 steps, $w = 7$	29.21	0.64	0.549	0.156

Table 15: CompBench of every saved checkpoint of the seed-0 118k runs. The last row is the weight average of the last five checkpoints.

Step	Caption visits	random	DINOv2 patches	VQAScore	patch Δ	VQAScore Δ
5,000	20,000	0.4570	0.4793	0.4607	+0.0223	+0.0037
10,000	40,000	0.4572	0.4539	0.4515	−0.0033	−0.0057
15,000	60,000	0.4237	0.4454	0.4463	+0.0217	+0.0226
20,000	80,000	0.4342	0.4687	0.4579	+0.0345	+0.0237
30,000	120,000	0.4600	0.4612	0.4704	+0.0013	+0.0104
40,000	160,000	0.4494	0.4706	0.4721	+0.0211	+0.0227
50,000	200,000	0.4567	0.4682	0.4684	+0.0115	+0.0117
56,974	227,896	0.4573	0.4587	0.4516	+0.0014	−0.0056
average of last 5		0.4642	0.4865	0.4827	+0.0222	+0.0185

Table 16: Optimizer batch size at 118k, seed 0. Batch 16 uses gradient accumulation over four captions per GPU at the same data budget. CompBench / GenEval2×100.

Batch (updates)	random, raw final	random, averaged	DINOv2 patches, raw final	DINOv2 patches, averaged
4 (56,974)	0.4573 / 19.51	0.4642 / 20.61	0.4587 / 20.18	0.4865 / 22.05
16 (14,244)	0.4684 / 20.28	0.4695 / 21.22	0.4796 / 20.90	**0.4916 / 22.68**
gap at batch 16, averaged	+0.0221 CompBench ($p = 4 \times 10^{-10}$), +1.45 GenEval2 ($p = 0.10$)			

10 Selection rule: argmax, Boltzmann weighting and sampling

Argmax selection is one point on a family of rules. The natural alternative weights the candidates by a Boltzmann distribution over the raw scores,

$$\pi_j = \frac{\exp(S_j/T)}{\sum_{l=1}^{4} \exp(S_l/T)}, \tag{12}$$

which recovers argmax as $T \to 0$ and uniform weighting as $T \to \infty$. Two estimators of the weighted objective $\sum_j \pi_j \mathcal{L}_j$ were trained, and they differ in cost. *Exact* weighting rolls out all four candidates and back-propagates each candidate's loss with weight π_j: four teacher rollouts and four student passes per caption, 2.3 times the wall-clock of the single-candidate trainer, at the same peak memory because candidates are back-propagated one at a time. *Sampled* weighting draws one candidate from π on every visit of the caption and trains on it alone, at single-candidate cost. The sampled estimator has the same expected gradient but a different target on each visit. Two controls complete the design: the random baseline, which draws once per caption and keeps that draw, and a *uniform-visit* arm that draws uniformly on every visit, which isolates target persistence from label quality.

10.1 Selection entropy and effective sample size

The temperature is calibrated on the scores themselves. Figure 3 and Table 17 summarise the DINOv2 patch score over the four candidates of a caption. Within a caption the scores have a standard deviation of 0.070 (range 0.18, margin between the best and the second-best candidate 0.05); between captions the mean score varies twice as much (0.14), which is why scores are only ever compared within a caption. The selection distribution of a caption is characterised by its normalised entropy $H(\pi)/\log 4$ and its effective sample size $\mathrm{ESS} = \exp H(\pi)$. At $T = 0.04$ the median ESS is 2.5 (mean entropy 0.62, mean weight of the argmax candidate 0.63); at $T = 0.08$ it is 3.3 (entropy 0.82). The 3k and 118k pools have the same statistics, so a temperature calibrated on one transfers to the other. Stronger teachers compress the candidates: on the 16-step $w = 4.5$ and 28-step caches the within-caption spread almost halves (0.041 and 0.038) and the entropy at a fixed temperature rises (0.75 and 0.76 at $T = 0.04$). This is the offline face of the vanishing selection gain of Section 11.3.

Table 17: DINOv2 patch score statistics per candidate cache ($N = 4$): within-caption standard deviation, mean range, mean margin between the two best candidates, and the mean normalised entropy and median effective sample size of the Boltzmann weights at the two temperatures used in training.

					$T = 0.04$		$T = 0.08$	
Cache (teacher)	captions	within std	range	top-2 margin	entropy	ESS	entropy	ESS
3k, 8 steps, $w = 7$	3,000	0.069	0.175	0.049	0.63	2.54	0.82	3.33
118k, 8 steps, $w = 7$	113,948	0.070	0.179	0.051	0.62	2.52	0.81	3.32
3k, 16 steps, $w = 4.5$	3,000	0.041	0.105	0.034	0.75	3.01	0.91	3.65
3k, 28 steps, $w = 7$	3,000	0.038	0.099	0.033	0.76	3.09	0.91	3.68

10.2 Result

All eight arms of Table 18 were trained on the 3k pool with the 3k configuration, three seeds each, from one frozen copy of the trainer, so every contrast in the table is a one-factor change. The random and argmax arms were retrained alongside the new ones. Their absolute level is 0.008 below the earlier runs of Table 6 (a different launch, three seeds each, within the seed

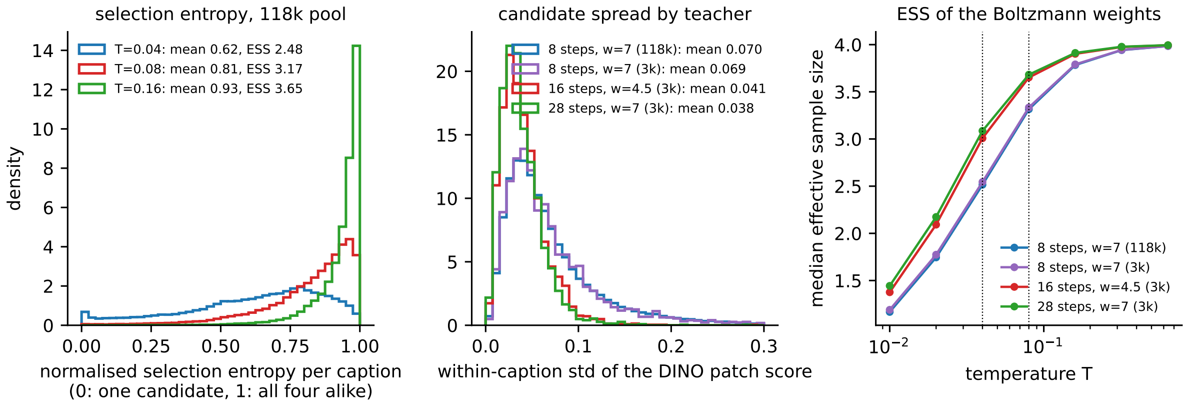

Figure 3: Left: distribution over captions of the normalised selection entropy at three temperatures (118k pool). Middle: within-caption spread of the DINOv2 patch score for the four caches; stronger teachers leave less to select on. Right: median effective sample size of the Boltzmann weights as a function of temperature; the two temperatures used in training are marked.

spread of 0.011), and the argmax gain replicates at +0.0203; contrasts should be read within this table.

Table 18: Selection rule, 3k pool, three seeds. CompBench and GenEval2×100 seed means; paired deltas against the random baseline and against argmax. *exact*: every candidate's loss weighted by π_j. *sampled*: one draw from π per visit. ESS is the median effective sample size of the weights.

Rule	Estimator	ESS	CompBench	Δ vs random (p)	Δ vs argmax (p)	GenEval2	Δ vs random (p)
random (fixed draw)	–	1	0.4499	–	−0.0203 (2×10^{-17})	21.12	–
argmax	–	1	**0.4702**	+0.0203 (2×10^{-17})	–	22.46	+1.35 (0.014)
Boltzmann $T = 0.04$	exact	2.5	**0.4703**	+0.0204 (4×10^{-17})	+0.0000 (0.98)	**22.65**	+1.53 (0.011)
Boltzmann $T = 0.08$	exact	3.3	0.4689	+0.0190 (3×10^{-15})	−0.0013 (0.56)	22.17	+1.06 (0.07)
uniform ($T \to \infty$)	exact	4	0.4557	+0.0058 (0.012)	−0.0145 (6×10^{-10})	21.28	+0.16 (0.77)
Boltzmann $T = 0.04$	sampled	2.5	0.4562	+0.0063 (0.009)	−0.0140 (3×10^{-9})	21.18	+0.07 (0.90)
Boltzmann $T = 0.08$	sampled	3.3	0.4517	+0.0019 (0.44)	−0.0185 (2×10^{-13})	22.37	+1.26 (0.03)
uniform-visit	sampled	4	0.4422	−0.0077 (0.002)	−0.0280 (10^{-27})	20.59	−0.52 (0.35)

Four conclusions follow.

- **Exact Boltzmann weighting ties argmax.** At $T = 0.04$ and $T = 0.08$ the exactly weighted objective matches argmax within 0.001 on CompBench and within the seed spread on GenEval2 ($p = 0.98$ and 0.56). There is no temperature at which soft selection beats hard selection, and the soft arm costs 2.3 times the compute. Argmax is the recipe.
- **The sampled estimator loses through target churn, not through the softer tilt.** At the same weights, sampling one candidate per visit is 0.0141 below exact weighting ($p = 3 \times 10^{-9}$), which is the whole of its deficit to argmax (0.0140). The uniform-visit control shows the mechanism directly: redrawing the target on every visit is 0.008 *worse* than a fixed random draw ($p = 0.002$). A consistency target that changes between visits of a caption costs more than a worse but stable target. The monotone decline of the sampled arms with temperature is a property of the estimator, not of soft selection.
- **The gain is weighting toward the best candidate, not exposure to more trajectories.** Uniform exact weighting trains on all four candidates of every caption, at four times the rollout cost, with a fixed target set and no selection. It is worth at most +0.006

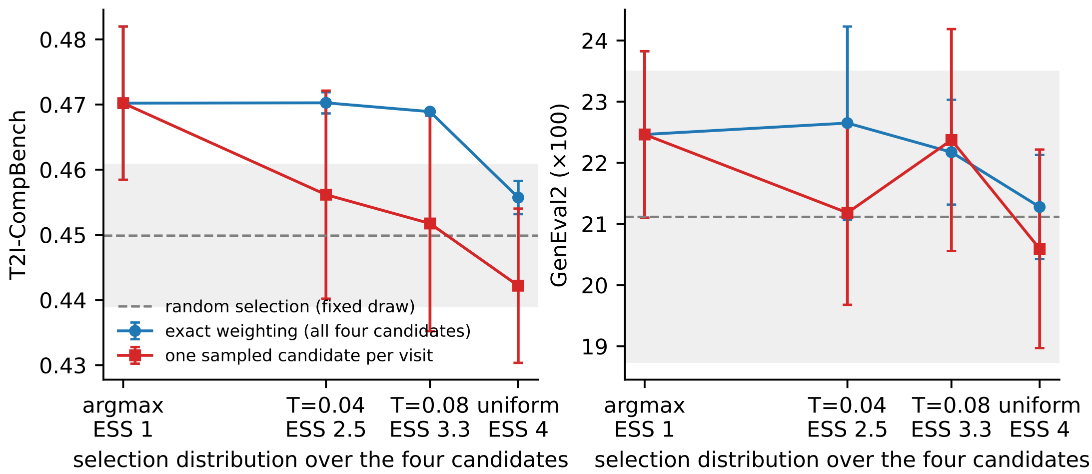

Figure 4: The two estimators of Boltzmann-weighted selection as a function of the effective sample size of the weights (three seeds, bars are seed standard deviations, grey band is the random baseline). Exact weighting is flat from argmax to ESS = 3.3 and drops only at uniform weighting; sampling one candidate per visit falls monotonically from argmax.

of the +0.020; the remaining +0.014 ($p = 6 \times 10^{-10}$ against argmax) comes from the weights.

- **Seed spread.** The exact-weighted arms have a seed standard deviation of 0.002 CompBench against 0.011 for the single-candidate arms: averaging the loss over four trajectories per caption also stabilises training, in line with the batch-size result of Section 9.5.

A decode-free variant that scores in latent space was not pursued. Decoding four candidates costs 61 ms per caption (Section 13), and the scorer needs the decoded image.

11 Ablations

All ablations use the 3k pool, three seeds and the 3k configuration unless stated.

11.1 Target horizon

The 1-step Tweedie target is compared with two targets that are closer to the trajectory endpoint: a 2-step chord, which uses z_{k+2} instead of z_{k+1}, and the endpoint z_8 itself. The 1-step target is the best for both arms and gives the largest selection gain (Table 19). The 1-step target is 27% off the endpoint at $k = 3$ and 10% at $k = 5$, so it is biased, but its lower variance across visits matters more.

Table 19: Target horizon. CompBench / GenEval2, seed means, and the selection gain of DINOv2 patches over random.

Target	random	DINOv2 patches	Selection gain (CompBench)
1-step Tweedie (used)	0.4584 / 20.73	0.4725 / 22.89	+0.0140
2-step chord	0.4533 / 21.09	0.4646 / 21.99	+0.0113 ($p = 3 \times 10^{-6}$)
endpoint z_8	0.4530 / 20.57	0.4565 / 20.97	+0.0035 (n.s.)

11.2 Objective: x_0 regression against segment-velocity imitation

A progressive-distillation-style objective places the student on state z_k and regresses the teacher's velocity chord to z_{k+1} with a mean-squared error. On the 8-step grid, deployed at 4 steps, it is much worse than the x_0 objective: random 0.4046 / 17.27 and VQAScore 0.4257 / 20.05 against 0.4584 / 20.73 and 0.4817 / 23.06. The selection gain persists under it (+0.021 CompBench), so selection does not depend on the objective, but the objective decides the absolute level.

11.3 Teacher quality

Two stronger teachers were tested with the same captions and seeds: 28 steps at $w = 7$ (candidates and training on the 28-step grid, supervised states 10, 14, 18, 22, 26, $\Delta \in \{3, \ldots, 10\}$), and 16 steps at the model-card guidance $w = 4.5$ (the 4-step inference grid nests in the 16-step grid; supervised states 6, 8, 10, 12, 14, $\Delta \in \{2, \ldots, 6\}$). Both compress the candidate pool a scorer can act on (Table 17): the oracle-minus-random VQAScore headroom falls from 0.0905 to 0.0415 (28 steps) and 0.0459 (16 steps), and the fraction of it the DINOv2 patch scorer recovers falls from 43.6% to 11.0% at 16 steps (top-1 agreement with the oracle 0.285, chance 0.25). The trained selection gain disappears (Table 20). A stronger teacher on its own is worth about as much on CompBench as scored selection on the 8-step teacher, and the two do not add. Guidance matters separately from step count: 16 steps at $w = 4.5$ lifts GenEval2 by 1.3 but CompBench by only 0.004 over the 8-step $w = 7$ teacher, while 28 steps at $w = 7$ lifts CompBench by 0.021.

Table 20: Teacher quality. CompBench / GenEval2, seed means over three seeds, and the paired selection gain within the same teacher.

Teacher	DINO headroom recovered	random	DINOv2 patches	Selection gain
8 steps, $w = 7$	43.6%	0.4584 / 20.73	0.4725 / 22.89	+0.0140 / +2.17
16 steps, $w = 4.5$	11.0%	0.4624 / 22.02	0.4664 / 22.16	+0.0040 ($p = 0.07$) / +0.14 ($p = 0.81$)
28 steps, $w = 7$	–	0.4797 / 23.06	0.4801 / 22.25	+0.0004 ($p = 0.87$) / −0.81 ($p = 0.18$)

Per teacher forward pass this is not a cost win for selection either: four 8-step candidates cost 32 passes per caption, one 28-step sample costs 28.

11.4 Regression onto the reference photograph

The reference photograph is the one sample we have from the distribution every scorer points at, so it was tested as a second regression target. On each update the photograph's latent is noised along its own straight path to two random supervised states, and the student's clean-latent estimate there is regressed onto the photograph latent with the same pseudo-Huber loss, added with weight $\lambda = 0.2$. On the 16-step teacher, three seeds, this term hurts both arms on both metrics: random $0.4624 \to 0.4281$ CompBench (-0.034, $p = 10^{-43}$, worst on UniDet at -0.052) and $22.02 \to 20.67$ GenEval2; DINOv2 patches $0.4664 \to 0.4407$ (-0.022) and $22.16 \to 20.32$. A single photograph per caption is a far noisier target than the teacher's trajectory, and it pulls the student off the trajectory it must reproduce at inference. The photograph is used for scoring only.

11.5 Evolving teacher

The frozen teacher never moves toward the high-score region; selection only filters what it produces. A self-distillation variant was therefore tested in which the teacher is an exponential moving average of the student. On every update the EMA teacher rolls out four fresh-seed candidates at $w = 1$, the DINOv2 patch scorer ranks them against the reference photograph on the fly, and the winner's EMA trajectory provides the targets. Runs start from the 118k averaged patch student (0.4865 / 22.05), use the 3k pool, 6,000 updates and one seed, and are compared with the same student continued against the frozen teacher (Table 21). Both the student and its EMA copy are evaluated.

Table 21: Evolving (EMA) teacher, seed 0, 6,000 updates from the 118k patch student. CompBench / GenEval2 of the raw student and of its EMA weights. $\bar{S}$ is the mean DINOv2 patch score of the teacher's fresh candidates at the end of training (0.60 at the start), H their normalised selection entropy at $T = 0.02$ (0.75 at the start).

Teacher	Selection	raw student	EMA weights	$\bar{S}$ at end	H at end
frozen (control)	DINOv2 patches, cached	0.4779 / 19.96	0.4847 / 20.68	–	–
EMA, decay 0.999	DINOv2 patches, online	0.1655 / 0.88	0.1789 / 2.09	0.03	0.93
EMA, decay 0.999	random	0.1546 / 1.00	0.1667 / 1.52	0.03	0.93
EMA, decay 0.999, frozen anchor 50%	DINOv2 patches, online	0.2095 / 6.28	0.2250 / 8.96	0.05	–
EMA, decay 0.9999	DINOv2 patches, online	0.4385 / 18.04	**0.4936 / 22.33**	0.62	0.75
EMA, decay 0.9999	random	0.4296 / 18.51	0.4917 / 22.03	0.61	–

A fast-tracking teacher (decay 0.999) collapses with or without selection and with a 50% frozen anchor: the candidates' score falls from 0.60 to 0.03 and their selection entropy rises from 0.75 to 0.93, so the scorer can no longer tell them apart, while the training loss falls and the student's samples turn into texture noise. The entropy of the online scores is an early warning of this collapse, visible thousands of updates before any image metric is computed. A slow teacher (decay 0.9999) is inert over 6,000 updates: the candidate score stays at 0.60 to 0.62, and selection within the slow loop is worth +0.002 CompBench and +0.3 GenEval2 against random ($p > 0.5$). Its EMA weights end 0.009 above the frozen-teacher control ($p = 0.01$), which is the effect of weight averaging, not of the moving teacher. Continuing the 118k student on the 3k pool with the frozen teacher lowers it (0.4865 to 0.4847), so the small pool is not a source of further gain either. The frozen teacher is kept.

11.6 Reference photograph aspect handling

Rescoring the 3k pool with the reference centre-cropped then resized, or resized then cropped, changes the top-1 pick on 12.2% and 11.8% of captions and lowers the recovered headroom from 43.6% to 41.0% and 40.9%. Cropping loses captioned objects at the edges, and mean-pooled patches tolerate the distortion of squashing. Squashing is therefore a design choice, not an oversight.

11.7 Patch mean against CLS token

The two DINOv2 readouts are the same model, the same decode, the same reference and the same normalisation; only the token summary differs. Offline, the patch mean recovers 43.6% of headroom against 39.9% for CLS. Trained, the patch arm is ahead by +0.0104 CompBench on identical prompts, positive on all three seeds, while the CLS arm's gain over random has an interval that includes zero. The CLS token is a global summary of what kind of image this is; the patch mean keeps object and layout structure, which is what compositional prompts test.

12 Qualitative examples

Figures 5 and 6 show prompts on which the scored student differs most from the random-selection student, chosen by the rule stated on each figure: the eight largest per-prompt margins, at most two per CompBench category, seed 0 of each arm, identical initial noise across all columns. They are cherry-picked by construction and show what the gain looks like where it is large, not typical behaviour. On CompBench the scored student wins 366 prompts, loses 229 and ties 1,803 within 0.1; on GenEval2 it wins 169, loses 140 and ties 491 of 800, and both students score below 0.05 on 27% of the prompts. The columns also show the base model at 4 steps with and without guidance, which fails, and the 28-step teacher, which the students approach. The recurring pattern in the wins is an object that the random-selection student drops or merges (the second object of a two-object prompt, the count of a numeracy prompt) and that the scored student renders, together with cleaner colour and material binding.

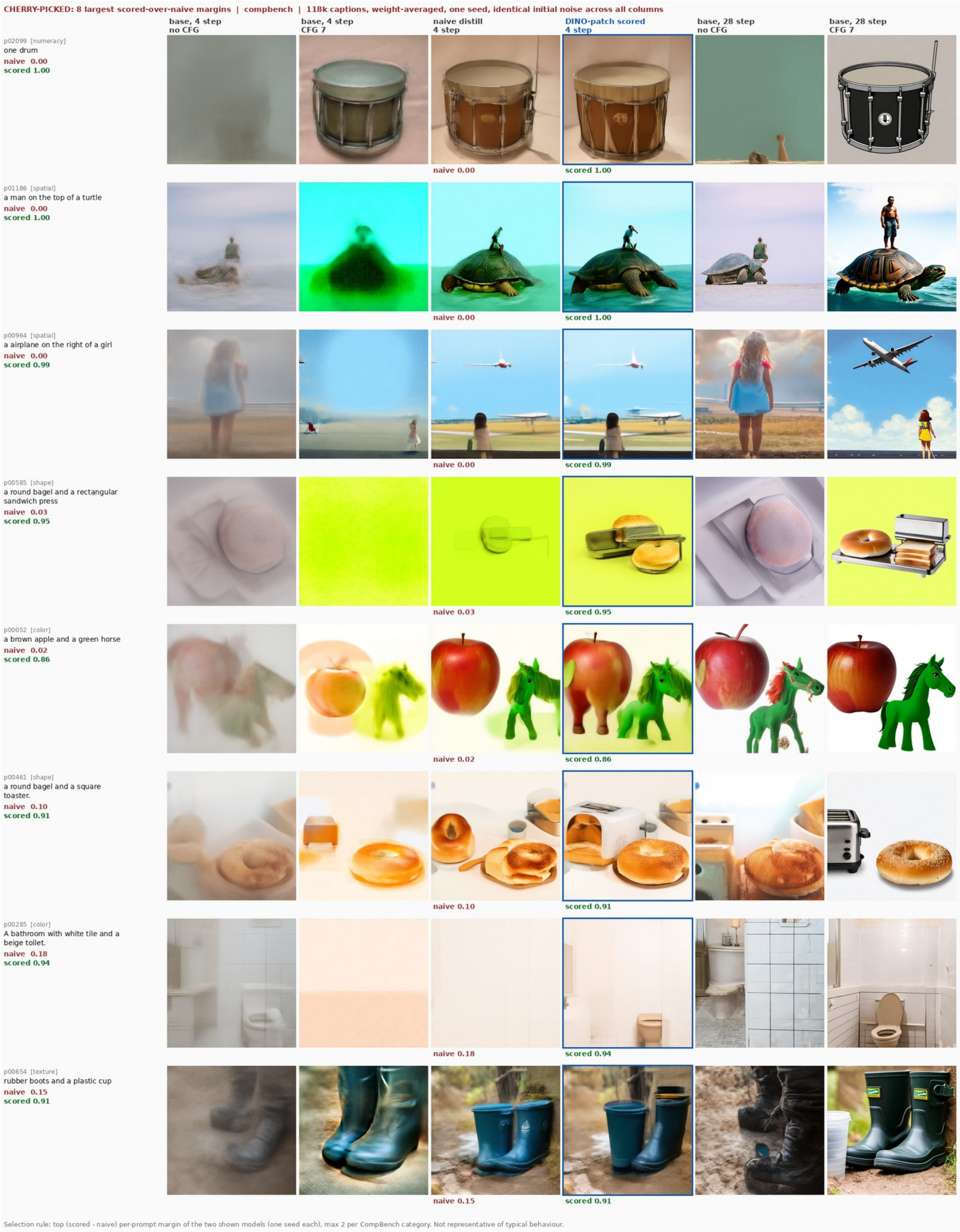

Figure 5: T2I-CompBench prompts with the largest scored-over-random margins at 118k. Columns: base model at 4 steps without and with guidance, random-selection student, DINOv2 patch student (framed), base model at 28 steps without and with guidance. Per-prompt scores under each student.

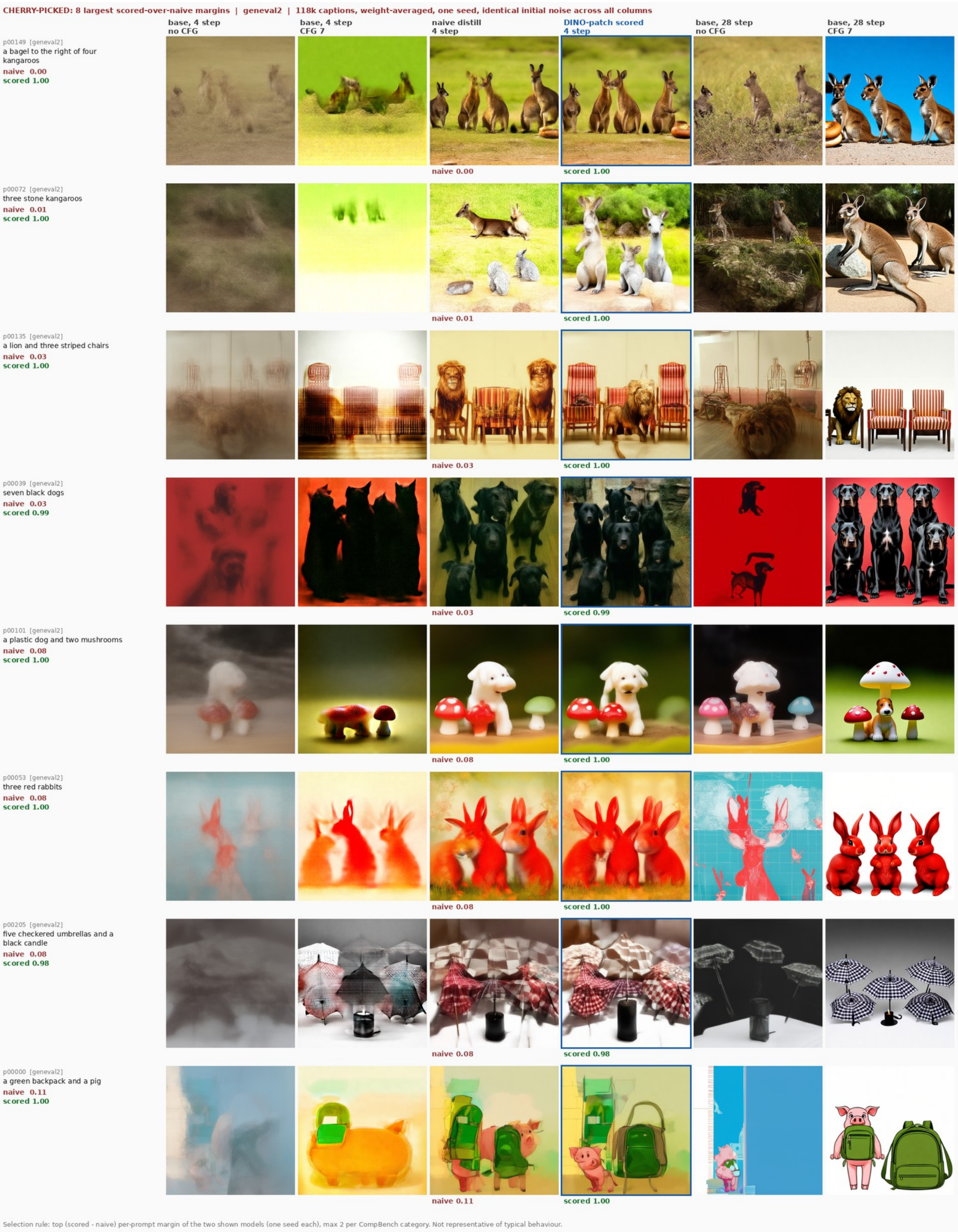

Figure 6: GenEval2 prompts with the largest scored-over-random margins at 118k, same layout as Figure 5.

13 Cost

Per training caption the candidate pool costs 64 teacher forward passes (4 candidates, 8 steps, two guidance branches), four VAE decodes and five DINOv2 forward passes. Measured on one H100 at 512×512, generation takes 789 ms per caption, the four decodes 61 ms, DINOv2 scoring 32 ms, CLIP-I 48 ms and VQAScore 280 ms. End to end, DINOv2 selection costs 0.95 s per caption and VQAScore selection 1.19 s; candidate generation dominates and no scorer avoids it. Training costs 16 teacher passes and five student forward-backward passes per caption. A 3k run takes 1.6 hours on one H200; the 118k run takes 12.5 hours on four H200s. The DINOv2 scorer needs no text model and no large VLM at selection time, which is its practical advantage over VQAScore, ImageReward and PickScore. The exact Boltzmann arm of Section 10, which rolls out and trains on all four candidates, takes 2.6 s per update against 1.15 s for the single-candidate trainer (3.5 h against 1.2 h for a 3k run) for no gain over argmax.

14 Design choices

- **Fixed teacher targets, no EMA.** Every target is a function of the frozen teacher's trajectory. There is no bootstrapping and no target drift. Consistency models use an EMA or stop-gradient student copy; we do not.
- **Hard selection, stored once.** One candidate per caption, chosen offline. The trainer rolls out one trajectory per caption for every arm, so compute is identical across arms. Exact Boltzmann weighting of all four candidates ties argmax at 2.3 times the cost, and drawing one candidate per visit from soft weights loses through target churn (Section 10). Four candidates are enough; eight add nothing (Section 9.6).
- **Guidance internalised.** The student trains with one conditional pass against guided teacher trajectories and is sampled at $w = 1$. This halves the student's inference cost relative to a guided sampler.
- **Re-roll instead of store.** Trajectories are re-rolled from integer seeds. The cache holds only seeds, scores and indices.
- **Reference photograph squashed to a square.** Cropping discards captioned content; see the aspect ablation.
- **Mean-pooled patches, CLS dropped.** The patch mean carries layout; the CLS token does not select well.
- **Weight averaging at scale.** The last five checkpoints are averaged because single checkpoints of a long run oscillate by more than the effect.
- **One sealed evaluation pool and fixed evaluation seeds.** Every arm and seed is scored on identical prompts and identical noise, so all contrasts are paired.

15 Limitations

- **The scorer needs a reference photograph.** DINOv2 selection is only possible where each caption has a real image. It cannot be applied to synthetic prompt sets without one.
- **Grids do not nest.** The 4-step inference grid shares only its endpoints with the 8-step training grid, so the student is deployed at $\sigma = 0.858$ and 0.602, which it never saw as training inputs, and is never trained at $\sigma = 0.009$ (the last step is scaled by 0.009 and is immaterial). Grids with $K \in \{7, 10, 13, 16, \dots\}$ nest exactly; this was not used here.

- **Duplicate student inputs.** With five independent Δ_k draws, 81% of updates contain at least one repeated input state and the expected number of distinct inputs is 3.9 of 5. This does not change the objective, only its variance, and wastes about 22% of student passes.
- **Seeds.** Every headline number is three training seeds. The single-seed contrasts (the VQAScore arm at 118k, the batch-size row, the evolving-teacher runs) are marked as such and resolve about 0.01 CompBench.
- **Checkpoint averaging is part of the method at scale.** Without it the 118k gap is +0.007 instead of +0.0175. Averaging is applied identically to every arm, but the reported 118k numbers are for averaged weights.
- **Selection gain depends on candidate variance.** With a 28-step or a 16-step $w = 4.5$ teacher the within-caption score spread halves and the gain vanishes. Scored selection is a fixed-teacher gain, not a substitute for teacher quality.
- **VQAScore is circular.** Its gains share a model family with the BLIP-VQA categories of CompBench and the VLM judge of GenEval2. It is included as a reference, not as the proposed method.
- **Metric properties.** GenEval2's geometric mean rewards partial credit, and CompBench 2D-spatial has dead prompts (Section 7.4). Both are constant across arms and do not reorder students, but they limit what absolute numbers mean.

16 Reproducibility

The release code is the `scored-consistency-distillation` branch of the project repository. It contains the pool builder, the candidate scorer, the trainer with the `random`, `dino_patch`, `boltzmann`, `boltzmann_sample` and `uniform_visit` selectors and gradient accumulation, the checkpoint averager, the evaluation scripts (CompBench with the one- and ten-image protocols, GenEval2, COCO VQAScore, FID, reference CMMD, precision and recall), the figure script and the LSF launch scripts. The release trainer was verified to be bit-identical to the experimental trainer that produced every number in this document, on all 909 parameter tensors after six optimizer updates under deterministic kernels, for every selector and for gradient accumulation, with an original-versus-original control at exactly zero. Every evaluation writes a pinned record with evaluator commits, package versions, pool hashes and the checkpoint hash, and fails hard on missing images. Training runs log per-state losses, selection statistics, throughput, sample grids and a source snapshot to Weights & Biases.

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